

Technology #154

Implement lead-free solder joint
reinforcement techniques that enhance
thermo-mechanical fatigue resistance without
adding significant mass or compromising heat
dissipation

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Implement lead-free solder joint reinforcement techniques that enhance thermo-mechanical fatigue resistance without adding significant mass or compromising heat dissipation

WHY DOES IT MATTER?

Documented aerospace reliability problem: BGA solder joints in satellite electronics accumulate thermo-mechanical fatigue damage from repeated thermal cycling between sunlight and eclipse (~-150C to +120C per orbit, ~5500 cycles/year in LEO), eventually cracking and producing intermittent or permanent open circuits. The failure is field-documented across aerospace electronics heritage programs and is a dominant driver of avionics derating and redundancy mass. This description is qualitative problem-domain context from the engineering literature; no specific incident statistics are asserted here (the engine's own retrieval must find and custody the evidence).

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 8 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in thermal. 9-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 9-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Accelerated thermal cycling testing (-150°C to +120°C) with and without the reinforcement technique, measuring crack initiation and propa.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: temperature-rise mapping at max duty. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden; interface contact resistances dominate at small scale and must be measured, not assumed zero; steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected); TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.; TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.; TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.; Model input variables requiring sourced values before any numeric result: h ($W/(m^2.K)$), A (m^2), dT (K), L_i (m), k_i ($W/(m.K)$), A_i (m^2), k ($W/(m.K)$), A (m^2), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , $W/(m.K)$] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m^2] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: measures to improve the reliability of solder joints

POTENTIAL VALUE DRIVER: Addresses: Documented aerospace reliability problem: BGA solder joints in satellite electronics accumulate thermo-mechanical fatigue damage from repeated thermal cycling between sunlight and eclipse ($\sim 150C$ to $+120C$ per orbit, ~ 5500 cycles/year in LEO), eventually cracking and producing intermittent or permanent open circuits. The failure is field-documented across aerospace electronics heritage programs and is a dominant driver of avionics derating and redundancy mass. This description is qualitative problem-domain context from the engineering literature; no specific incident statistics are asserted here (the engine's own retrieval must find and custody the evidence).

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: thermal

Disciplines: heat transfer, thermal engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: Implement lead-free solder joint reinforcement techniques that enhance thermo-mechanical fatigue resistance without adding significant mass or compromising heat dissipation

Subsystems:

?: heat source characterization - ?

?: conduction path / spreader - ?

?: reject interface - ?

?: safety limit interlock - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Implement lead-free solder joint reinforcement techniques that enhance thermo-mechanical fatigue resistance without adding significant mass or compromising heat dissipation

Key Physics: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: CONDITIONAL; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: NOT ESTABLISHED.

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected). Engaged variables: k .

Assumptions:

- h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden
- interface contact resistances dominate at small scale and must be measured, not assumed zero
- steady-state conduction (no transient term)
- isotropic, homogeneous medium conductivity k
- one-dimensional gradient (lateral gradients neglected)
- TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.
- TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.
- TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.

- Model input variables requiring sourced values before any numeric result: h (W/(m².K)), A (m²), dT (K), L_i (m), k_i (W/(m.K)), A_i (m²), k (W/(m.K)), A (m²), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , W/(m.K)] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m²] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'solder joint fatigue cracking under thermal cycling', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'hotspot exceeding material/safety limit', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'interface delamination raising resistance over life', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coolant/flow path blockage (if active)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'thermal runaway coupling to electronics', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'condensation at cold surfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Any intervention must survive qualification-level thermal cycling (-150C to +150C) REFACT Add more than a few grams	MODELLING
DI-002	Mechanism driver (proposed)	measures to improve the reliability of solder joints	MODELLING
DI-003	Expected effect target	Reduced thermo-mechanical fatigue damage in BGA solder joints during the MODELLING , extending service life and redu	MODELLING
DI-004	Domain design input pattern: dissipation applicable by SYSTEM via tokens ['dissipation']	invention applicable by SYSTEM via tokens ['dissipation']	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: ambient and applying by NOT ESTABLISHED (candidate pattern; confirm again)	invention applicable by NOT ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: safety limit applicable by NOT ESTABLISHED (candidate pattern; confirm again)	invention applicable by NOT ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidate standards (e.g. IEC 60601-1) (general safety, UNKNOWING) NOT ESTABLISHED (candidate pattern; confirm again)	invention applicable by NOT ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED
DI-008	Custodied observation: Reliability issues of the Europe solder joints in the NOT ESTABLISHED (candidate pattern; confirm again)	invention applicable by NOT ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	heat source characterization realizing Implement lead-free solder joint reinforcement techniques that enhance thermo-mechanical fatigue resistance with	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-002	conduction path / spreader realizing Implement lead-free solder joint reinforcement techniques that enhance thermo-mechanical fatigue resistance witho	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-003	reject interface realizing Implement lead-free solder joint reinforcement techniques that enhance thermo-mechanical fatigue resistance without adding sig	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-004	safety limit interlock realizing Implement lead-free solder joint reinforcement techniques that enhance thermo-mechanical fatigue resistance without addin	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-005	heat source characterization interfaces with conduction path / spreader interfaces with reject interface interfaces with safety limit interlock: signal/en	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-006	conduction path / spreader interfaces with reject interface interfaces with safety limit interlock: signal/en	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-007	reject interface interfaces with safety limit interlock: signal/en	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data points required by system to test the solder joint reinforcement tech	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-009	Flow/force/signal path geometry realizing the mechanism holds its stated behavior computed bounds for heat generation at source; verification o	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-010	Operating range for heat generation at source within which the mechanism holds its stated behavior computed bounds for spreader conductivity; verification o	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-011	Operating range for spreader conductivity within which the mechanism holds its stated behavior computed bounds for interface contact resistance; verification	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-012	Operating range for interface contact resistance within which the mechanism holds its stated behavior computed bounds for reject-path area; verification of range e	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-013	Operating range for reject-path area within which the mechanism holds its stated behavior computed bounds for allowable temperature rise; verification	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-014	Operating range for allowable temperature rise within which the mechanism holds its stated behavior computed bounds for ambient worst case; verification of rang	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-015	Operating range for ambient worst case within which the mechanism holds its stated behavior computed bounds for ambient worst case; verification of rang	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-016	No closed-loop control is proposed: the invention operates in open-loop as specified; this record exists so the omission is explicit and auditable	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-017	Information path: sensing element -> decision/monitor CONCEPTUAL (act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-018	Fixture enabling: temperature-rise mapping at max duty CONCEPTUAL (act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-019	Fixture enabling: interface resistance characterization CONCEPTUAL (act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-020	Fixture enabling: worst-case ambient soak test — rig CONCEPTUAL (act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']
DO-021	Fixture enabling: temperature-rise mapping at maximum duty CONCEPTUAL (act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPTUAL	invention applicable by SYSTEM via tokens ['dissipation']

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
heat generation at source	UNKNOWN (no sourced value)	W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
spreader conductivity	UNKNOWN (no sourced value)	W/(m.K)	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
interface contact resistance	UNKNOWN (no sourced value)	K/W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
reject-path area	UNKNOWN (no sourced value)	m^2	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
allowable temperature rise	UNKNOWN (no sourced value)	K	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
ambient worst case	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
solder joint fatigue cracking	Failure by solder joint fatigue	design control the thermal stress by SH-2 (excellent design)	monitors the invents, triggered failure	operating (Ang. d
hotspot exceeding material safety limit	Failure by hotspot exceeding material safety limit	spread safety limit; heat-seal spreading gutters; heat sink with high temperature	at the hotspot modes (AST	
interface delamination	Failure by interface delamination	Massive resistance over life cycle; de-impedance with high SW gaps and high candidate failure modes (San		
coolant/flow path blockage	Failure by coolant/flow path blockage (if active)	particulate in middle of block up a link showing channels; candidate failure modes (Ad. t		
thermal runaway coupling	Failure by thermal runaway coupling	policy of thermal management; analysis; raised leakage in KGSW, which raise the candidate failure mode (ed b		
condensation at cold surfaces	Failure by condensation at cold surfaces	low power surfaces operating at low power point only; provide cooling in the SW; raised design candidate failure and de (ode		

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
solder joint fatigue cracking	Failure by solder joint fatigue cracking under thermal cycling	designer controls NOT ESTABLISHED	Failure by solder joint fatigue cracking under thermal cycling
hotspot exceeding material safety limit	Failure by hotspot exceeding material/safety limit	designer controls NOT ESTABLISHED	Failure by hotspot exceeding material/safety limit
interface delamination raising risk of system failure	Failure by interface delamination raising risk of system failure	designer controls NOT ESTABLISHED	Failure by interface delamination raising risk of system failure
coolant/flow path blockage	Failure by coolant/flow path blockage (if active)	designer controls NOT ESTABLISHED	Failure by coolant/flow path blockage (if active)
thermal runaway coupling	Failure by thermal runaway coupling to electrical system	designer controls NOT ESTABLISHED	Failure by thermal runaway coupling to electrical system
condensation at cold surfaces	Failure by condensation at cold surfaces: surface is below the operating point	designer controls NOT ESTABLISHED	Failure by condensation at cold surfaces: surface is below the operating point

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Accelerated thermal cycling testing (-150C to 120C) with cycling testing	Accelerated thermal cycling testing (-150C to 120C) with cycling testing	Accelerated thermal cycling testing (-150C to 120C) with cycling testing	Accelerated thermal cycling testing (-150C to 120C) with cycling testing
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty
VF-004	temperature-rise mapping at maximum	temperature-rise mapping at maximum	temperature-rise mapping at maximum	temperature-rise mapping at maximum
VF-005	thermal-resistance trend over accelerated cycling	thermal-resistance trend over accelerated cycling	thermal-resistance trend over accelerated cycling	thermal-resistance trend over accelerated cycling
VF-006	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty
VF-007	worst-case ambient soak with staged power	worst-case ambient soak with staged power	worst-case ambient soak with staged power	worst-case ambient soak with staged power

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	biocompatible thermal spreader (graphite/AlN)	graphite/AlN	electronics thermal practice	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
biocompatible thermal spreader	biocompatible thermal spreader	1	STANDARD	biocompatible thermal spreader	Not Established	STANDARD	Stability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- bonded spreader + encapsulation: domain registry manufacturing route - candidate only; no process qualification exists
- thermal interface material application control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter heat generation at source [q W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter spreader conductivity [k_i W/(m.K)]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter interface contact resistance [R_total K/W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter reject-path area [A m^2]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter allowable temperature rise [dT K]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter ambient worst case [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}

- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification	1050 Hz frequency response	NOT ESTABLISHED	1050 Hz frequency response	1050 Hz frequency response
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	constraint-compliance	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Accelerated thermal cycling	NOT ESTABLISHED	Accelerated thermal cycling	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment	NOT ESTABLISHED	run killer experiment	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	temperature-rise mapping	NOT ESTABLISHED	VF-003 verification	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	temperature-rise mapping	NOT ESTABLISHED	VF-004 verification	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	thermal-resistance trend	NOT ESTABLISHED	VF-005 measurement	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	temperature-rise mapping	NOT ESTABLISHED	VF-006 verification	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	worst-case ambient	NOT ESTABLISHED	VF-007 verification	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 9-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.